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LEARNER GUIDE

For

Agentic AI for Email Marketing Campaign

TGS Ref No: TGS-2026064473

Conducted by

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Version 1.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	24 August 2026	Revamped n8n and MailerLite edition: consented lead acquisition, multi-agent drafting, human approval, campaign scheduling, analytics and controlled optimization.	Dr Alfred Ang

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Course Overview

This guide supports the 2-day, 16-hour WSQ course Agentic AI for Email Marketing Campaign (TGS-2026064473). It is the procedural companion to the concept-led slide deck. The six labs form one continuous campaign system: acquire a consented lead, produce a controlled newsletter, schedule the approved version in MailerLite, measure outcomes and feed evidence into the next campaign brief.

Learning Outcomes

- LO1: Implement email marketing campaigns in alignment to strategic marketing objectives.
- LO2: Analyse email marketing campaign outcomes using sales performance data and consumer insights.
- LO3: Analyse market trends and customer reviews.

Assessment Alignment

- Written Assessment (WA) - six open-ended Short Answer Questions, 60 minutes, open book.
- Practical Performance (PP) - five connected workflow tasks, 60 minutes, open book.
- Question papers preserve the original 6 K-coded WA questions and 5 A-coded PP tasks.

Security boundary: Use sample leads and a non-production MailerLite group. Store tokens only in n8n Credentials. Remove PII and credential values from screenshots and exported evidence.

Environment Setup

1. Sign in to a training n8n workspace and create a folder for this course.
2. Create a MailerLite training group and note its group ID. Do not use a production audience.
3. Create an n8n Header Auth credential for MailerLite. Set the header name to Authorization and the value to Bearer followed by the API token.
4. Create or select a verified training sender in MailerLite. Record its email address without exposing any verification token.
5. Download the labs folder. Confirm each of the six lab folders contains README.md, CHECKLIST.md, an importable workflow JSON and sample data.
6. Import only the workflow required for the current lab and run it first with pinned or supplied sample data.

MailerLite response rule: The subscriber upsert can return HTTP 201 for a new subscriber or HTTP 200 for an existing subscriber. Both are valid when the returned identity and status match the request.

Topic 01 - Implementing Agentic AI Email Marketing Campaigns

Consent -> brief -> specialist agents -> approval -> MailerLite schedule

Concept model

- Translate strategy into a machine-readable campaign brief and acceptance contract.
- Separate audience, research, drafting and compliance roles across agent boundaries.
- Gate every external send or schedule action with deterministic human approval.
- Treat consent, unsubscribe state, secrets and audit evidence as first-class data.

Lab 1 - Consent-First Lead Capture to MailerLite

Duration	75 minutes
Alignment	LO1 K3 A1
Tools	n8n Form Trigger, Code, IF, HTTP Request, MailerLite API
Deliverable	A production-shaped lead form, MailerLite upsert, rejection path and evidence record.

Build an n8n form workflow that validates explicit consent, normalises a lead, upserts the subscriber into a MailerLite group and preserves audit evidence.

How the workflow works

- Form Trigger produces a structured submission
- Code normalises and validates fields
- IF blocks absent consent
- HTTP Request upserts the subscriber
- Evidence node stores safe response fields

Contract to preserve

- email is trimmed and lowercased
- consent must equal true
- consent_text_version is recorded
- group_id is configuration, not model output
- API token remains in an n8n credential

Detailed step-by-step procedure

7. Open n8n and create a new workflow named LAB01 - Consent-First Lead Capture.
8. Import lead-capture-mailerlite.json from this lab folder.
9. Open the Form Trigger and review the Name, Email, Company and explicit consent fields.
10. Compare the displayed consent wording with consent_text_version in the Normalize Lead node.
11. Create an n8n Header Auth credential for MailerLite; use Authorization with value Bearer plus your API token.
12. Attach the credential to the Upsert MailerLite Subscriber HTTP Request node; never paste the token into a node field.
13. Replace TRAINING_GROUP_ID in the Workflow Configuration node with a MailerLite group ID created for training.
14. Run the workflow in test mode and open the Form Trigger Test URL.
15. Submit a valid sample lead with consent checked and capture the Normalize Lead output.
16. Confirm the IF node routes the item to the approved branch only when consent is true.
17. Inspect the HTTP request body and verify email, fields, groups and opted_in_at match the contract.
18. Verify MailerLite returns 201 for a new subscriber or 200 for an existing subscriber.
19. Submit the same email again with an updated company and confirm the upsert is non-destructive.
20. Submit a row from sample-leads.csv with consent false and confirm no MailerLite call executes.
21. Export the execution evidence without the API token and complete CHECKLIST.md.

Acceptance criteria

- valid consent reaches MailerLite

- missing consent never calls MailerLite
- subscriber ID/status captured
- credential value absent from exports

Evidence to capture

- test submission
- normalised payload
- MailerLite subscriber ID and status
- n8n execution ID

Troubleshooting and failure handling

- invalid email returns a controlled form response
- unsubscribed status is never silently overridden
- 200 and 201 are both accepted
- 429 follows bounded exponential backoff

Working files: Open the matching folder under labs/lab-01-*. Complete its README procedure and CHECKLIST.md. Workflow exports must contain no credentials.

Lab 2 - Campaign Brief and Audience Contract

Duration	60 minutes
Alignment	LO1 K1-K2
Tools	n8n Form Trigger, Edit Fields, Code, Structured Output
Deliverable	A campaign-brief intake form, JSON schema validator and audience/message contract.

Turn a marketing objective into a validated campaign brief that every specialist agent consumes consistently.

How the workflow works

- Brief intake captures objective and constraints
- Validator enforces required fields
- Audience contract defines inclusion and exclusion
- Versioned brief becomes agent context

Contract to preserve

- one primary objective
- one primary CTA
- eligible group and exclusions
- brand voice and prohibited claims
- evidence source IDs
- schedule and timezone

Detailed step-by-step procedure

22. Import campaign-brief-validator.json into n8n and open the Workflow Configuration node.

23. Open campaign-brief.json and read the objective, offer, audience, exclusions, voice, claims and schedule fields.
24. Set the campaign name and choose one measurable primary objective.
25. Define the eligible MailerLite group and at least two audience exclusions.
26. Write one primary CTA and the destination URL; do not use multiple competing CTAs.
27. Add brand voice attributes and prohibited tone or claim patterns from brand-guidelines.md.
28. List the approved evidence source IDs that agents may cite.
29. Set the intended schedule in local time and provide the explicit timezone identifier.
30. Execute the validator and inspect each validation_error item.
31. Remove the audience exclusions temporarily and verify the workflow fails closed.
32. Restore the exclusions, increment brief_version and execute again.
33. Confirm the final output contains a stable campaign_id and brief hash.
34. Save the approved brief as evidence for Lab 3.

Acceptance criteria

- all required fields pass
- exclusions present
- one CTA only
- brief version and hash generated

Evidence to capture

- campaign-brief.json
- validator result
- audience inclusion/exclusion table
- brief version hash

Troubleshooting and failure handling

- ambiguous objectives fail validation
- missing exclusions stop the pipeline
- unversioned edits create a new brief version
- PII is not inserted into prompts

Working files: Open the matching folder under labs/lab-02-*. Complete its README procedure and CHECKLIST.md. Workflow exports must contain no credentials.

Lab 3 - Multi-Agent Newsletter Editorial System

Duration	120 minutes
Alignment	LO1 K1-K2 A2
Tools	n8n AI Agent, OpenAI Chat Model, Structured Output Parser, Loop Over Items, Code
Deliverable	An importable multi-agent n8n workflow that produces a grounded newsletter draft with quality scores and provenance.

Coordinate strategist, researcher, copywriter, critic and refiner agents through explicit schemas and a bounded revision loop.

How the workflow works

- Strategist produces the content plan
- Researcher maps claims to source IDs
- Copywriter emits structured content
- Critic scores mandatory dimensions
- Refiner loops until threshold or escalation

Contract to preserve

- structured JSON output
- maximum two revisions
- every factual claim carries a source ID
- critic minimum score 80 on every dimension
- no external send tool attached

Detailed step-by-step procedure

35. Import multi-agent-newsletter.json and connect an approved n8n chat-model credential.
36. Open every AI Agent node and read its role, allowed inputs and required output.
37. Confirm the Workflow Configuration node points to the approved Lab 2 campaign brief.
38. Review approved-sources.md and add one current, authoritative source relevant to the campaign.
39. Run the workflow once with pinned sample data before using a live model.
40. Execute the Strategist Agent and verify it returns audience need, narrative arc and one CTA.
41. Execute the Research Agent and verify every proposed factual claim has a source_id.
42. Execute the Copywriter Agent and validate its output against newsletter-output-schema.json.
43. Inspect the Critic Agent scorecard for brand, evidence, compliance and actionability.
44. Force one low compliance score in pinned data and confirm the Refiner path executes.
45. Confirm revision_count never exceeds the configured maximum of two.
46. Confirm a still-failing draft is marked needs_human_review rather than silently accepted.
47. Inspect the final HTML and plain-text alternatives for the same message and CTA.
48. Save the final content JSON and critic scorecard for Lab 4.

Acceptance criteria

- five distinct roles visible
- source IDs preserved
- schema valid
- bounded revision loop
- no send/schedule action

Evidence to capture

- agent outputs
- critic scorecard
- revision count
- final content JSON
- execution graph

Troubleshooting and failure handling

- parser errors route to repair
- missing source IDs force revision
- loop limit escalates to human

- agent outputs never overwrite raw brief

Working files: Open the matching folder under labs/lab-03-*. Complete its README procedure and CHECKLIST.md. Workflow exports must contain no credentials.

Lab 4 - Human Approval and MailerLite Scheduling

Duration	105 minutes
Alignment	LO1 K4 A3
Tools	n8n Chat Trigger, AI Agent, Chat HITL Tool, HTTP Request, Code
Deliverable	A human-review gate, payload hash check, MailerLite draft creation and schedule transaction with recovery controls.

Bind a human decision to the exact newsletter payload, create a MailerLite draft and schedule only the approved version.

How the workflow works

- Draft payload is canonicalised and hashed
- HITL presents subject, audience, CTA and schedule
- Reviewer approves, rejects or requests revision
- Hash is rechecked
- MailerLite draft and schedule APIs run

Contract to preserve

- authenticated reviewer
- approval expiry
- payload hash equality
- verified sender
- group ID allow-list
- future schedule with timezone

Detailed step-by-step procedure

49. Import hitl-mailerlite-scheduler.json and connect the same approved chat-model credential used in Lab 3.
50. Create an n8n Header Auth credential for MailerLite and attach it to both HTTP Request nodes.
51. Open approved-newsletter.json and replace placeholder sender, group and timezone values with training values.
52. Confirm the verified sender already exists in MailerLite.
53. Execute Canonicalize and Hash and record the payload_hash.
54. Open the Review Newsletter tool and confirm it presents subject, audience group, CTA URL, schedule and hash.
55. Run the workflow and choose Reject; verify neither MailerLite node executes.
56. Run again, choose Revise or deny, and confirm reviewer feedback is stored with the content version.
57. Run again, choose Approve, then change one character in the content before the hash check.
58. Verify the hash mismatch blocks the external write and returns the draft to review.
59. Restore the approved content and approve again.

60. Inspect Create Campaign Draft and confirm the response contains a campaign ID and draft status.
61. Inspect Schedule Campaign and verify the schedule is in the future with the intended timezone.
62. Run the recovery check and confirm it reads campaign status before any retry.
63. Capture the decision, campaign ID and scheduled timestamp as assessment evidence.

Acceptance criteria

- reject path makes no write
- hash mismatch blocks
- approved draft created
- campaign scheduled once
- evidence contains reviewer and campaign ID

Evidence to capture

- review request
- decision payload
- hash comparison
- MailerLite campaign ID
- scheduled timestamp

Troubleshooting and failure handling

- hash mismatch blocks write
- rejection returns feedback
- 422 exposes safe validation detail
- retry checks campaign status before a second write

Working files: Open the matching folder under labs/lab-04-*. Complete its README procedure and CHECKLIST.md. Workflow exports must contain no credentials.

Topic 02 - AI-Driven Campaign Performance and Consumer Insights Analysis

MailerLite events -> normalised facts -> KPI model -> diagnostic dashboard

Concept model

- Model campaign, delivery and subscriber events at the correct grain.
- Distinguish delivery, engagement and business outcomes in a metric tree.
- Use cohort and funnel views to diagnose where performance is lost.
- State limitations of open tracking, attribution and small samples.

Lab 5 - Newsletter Analytics Collector and Dashboard

Duration	120 minutes
Alignment	LO2 K5 A4
Tools	n8n Webhook, Schedule Trigger, Code, HTTP Request, Respond to Webhook

Deliverable	A webhook/poll collector, event fact table, KPI calculation and HTML dashboard with target, cohort and funnel views.
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Ingest MailerLite events, normalise and deduplicate them, calculate metrics at defined denominators and publish a self-contained dashboard.

How the workflow works

- Fast webhook acknowledgement
- HMAC verification and deduplication
- normalised event fact
- metric aggregation
- HTML dashboard response

Contract to preserve

- event_id unique
- UTC timestamps
- metric denominators labelled
- cohort filters explicit
- PII excluded from dashboard

Detailed step-by-step procedure

64. Import analytics-dashboard.json and open the workflow notes that separate ingestion from presentation.
65. Open campaign-events.csv and identify the event grain, campaign key, subscriber key and timestamp.
66. Pin the first ten CSV rows at the Normalize Events node for a credential-free test.
67. Execute Normalize Events and verify UTC timestamps, allowed event types and source hashes.
68. Run the Deduplicate Events node twice and confirm the second run adds no duplicate facts.
69. Open metric-dictionary.md and confirm each metric denominator matches the Code node.
70. Execute Aggregate KPIs and manually recompute one delivery rate and one CTR.
71. Inspect the cohort output for new leads, existing customers and dormant subscribers.
72. Open the dashboard Webhook Test URL and verify the KPI cards, funnel and trend views render.
73. Select one campaign and one cohort and verify the dashboard labels reflect the filter.
74. Introduce one duplicate click event and confirm unique-click metrics do not double-count it.
75. Introduce a campaign with zero delivered events and confirm rates show N/A rather than a division error.
76. Compare actual results with targets and identify the largest material gap.
77. Write one evidence-based diagnosis and one controlled corrective action.
78. Export dashboard-sample.html and complete CHECKLIST.md.

Acceptance criteria

- events normalised
- duplicates ignored
- metrics recomputable
- dashboard renders
- diagnosis cites cohort/funnel evidence

Evidence to capture

- sample event payloads
- deduplication result
- metric dictionary
- dashboard HTML or screenshot
- diagnostic narrative

Troubleshooting and failure handling

- invalid signature rejected
- duplicate event ignored
- late event updates the correct window
- empty denominator returns null not infinity

Working files: Open the matching folder under labs/lab-05-*. Complete its README procedure and CHECKLIST.md. Workflow exports must contain no credentials.

Topic 03 - AI Agent Analysis of Market Trends and Customer Feedback

Signals -> evidence -> themes -> confidence -> controlled optimization

Concept model

- Separate observed signals from agent inference and recommended action.
- Combine quantitative trends with review themes and unsubscribe reasons.
- Use confidence, materiality and reversible-test thresholds before changing a campaign.
- Write approved learnings back into the next campaign brief without contaminating raw data.

Lab 6 - Trend and Customer Feedback Optimization Agent

Duration	90 minutes
Alignment	LO3 K6 A5
Tools	n8n Code, AI Agent, Structured Output Parser, IF, Chat HITL Tool
Deliverable	A trend/feedback analysis workflow with theme counts, confidence, counter-evidence, priority score and experiment contract.

Combine campaign trends with de-identified feedback, require evidence cards and produce a reversible experiment recommendation for human approval.

How the workflow works

- Load bounded performance window
- De-identify feedback
- Agent clusters themes
- Code checks evidence and calculates priority
- Human accepts experiment contract

Contract to preserve

- no raw email addresses in prompts
- theme counts link to source IDs
- confidence includes reason
- counter-evidence mandatory
- experiment has guardrail metrics

Detailed step-by-step procedure

79. Import trend-feedback-agent.json and inspect the analysis window and minimum-sample configuration.
80. Open customer-feedback.csv and identify which columns contain direct identifiers.
81. Execute De-identify Feedback and confirm email addresses and names are absent from the agent input.
82. Run the theme classifier with pinned sample data and inspect source_ids for every theme.
83. Verify the structured output contains finding, sample_size, window, confidence, counter_evidence and source_ids.
84. Force one finding to omit source_ids and confirm the evidence validator rejects it.
85. Compute the six-week CTR trend and compare it with the baseline in the workflow.
86. Combine the trend with feedback-theme counts; distinguish observed facts from inference.
87. Calculate priority for at least two improvement options using impact, confidence, reversibility and effort.
88. Select the higher-priority reversible test and define a single primary metric.
89. Add unsubscribe, complaint and bounce rates as guardrail metrics.
90. Define control/treatment allocation, analysis window and stop rule.
91. Submit the experiment contract to the human review tool and inspect approve/reject behavior.
92. Save the approved experiment contract and one rejected unsupported recommendation as evidence.

Acceptance criteria

- PII removed
- themes trace to sources
- trend has baseline/window
- counter-evidence present
- experiment has guardrails and approval

Evidence to capture

- de-identified feedback
- theme table
- trend chart
- evidence card
- approved experiment contract

Troubleshooting and failure handling

- insufficient sample returns monitor
- untraceable claim fails schema
- PII detector blocks prompt
- irreversible change requires escalation

Working files: Open the matching folder under labs/lab-06-*. Complete its README procedure and CHECKLIST.md. Workflow exports must contain no credentials.

Assessment Preparation

Use the following traceability map to prepare. The assessment tests the same contracts, workflows and evidence that appear in the deck and labs.

Criterion	What to demonstrate	Primary source
K1	Agentic campaign architecture, roles, boundaries and control points	Deck technical core and Lab 3
K2	Audience, message and brand contracts for newsletter generation	Lab 2 and Lab 3
K3	Consent-first lead capture and MailerLite subscriber lifecycle	Lab 1
K4	Campaign draft, approval and scheduling contracts	Lab 4
K5	Performance metrics, event data and dashboard interpretation	Lab 5
K6	Trend, feedback and continuous-improvement decision rules	Lab 6
A1	Build a consented lead-capture workflow that upserts MailerLite subscribers	Lab 1 evidence package
A2	Configure a multi-agent newsletter drafting and critique workflow	Lab 3 evidence package
A3	Apply a human approval gate and create/schedule a MailerLite campaign	Lab 4 evidence package
A4	Build and interpret a newsletter analytics dashboard	Lab 5 evidence package
A5	Analyse trends and customer feedback and recommend campaign changes	Lab 6 evidence package

Technical References

- **n8n AI Agent node:** <https://docs.n8n.io/integrations/builtin/cluster-nodes/root-nodes/n8n-nodes-langchain.agent/>
- **n8n Human-in-the-loop for tool calls:** <https://docs.n8n.io/advanced-ai/human-in-the-loop-tools/>
- **n8n Form Trigger node:** <https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.formtrigger/>
- **MailerLite Subscribers API:** <https://developers.mailerlite.com/api/subscribers>
- **MailerLite Campaigns API:** <https://developers.mailerlite.com/api/campaigns>
- **MailerLite Webhooks API:** <https://developers.mailerlite.com/api/webhooks>